

Kidney Biopsy

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Percutaneous kidney biopsy was first described in the early 1950s by Iversen and Brun¹ and Alwall.² These early biopsies were performed with the patient in the sitting position by use of a suction needle and intravenous urography for guidance. An adequate tissue diagnosis was achieved in less than 40% of these early cases. In 1954 Kark and Muehrcke³ described a modified technique using the Franklin-modified Vim-Silverman needle, with the patient in a prone position and an exploring needle used to localize the kidney before insertion of the biopsy needle. These modifications yielded a tissue diagnosis in 96% of cases, and no major complications were reported. Since then, the kidney biopsy procedure has remained largely unchanged, although the use of real-time ultrasound and refinement of biopsy needle design have offered significant improvements. Kidney biopsy now provides a tissue diagnosis in more than 95% of patients, with a life-threatening complication rate of less than 0.1%.

INDICATIONS FOR KIDNEY BIOPSY

Ideally, analysis of a kidney biopsy sample should identify a specific diagnosis, reflect the level of disease activity, and provide information to allow informed decisions about treatment. Although not always able to fulfill these criteria, the kidney biopsy remains a valuable clinical tool and is particularly beneficial in the clinical situations discussed in [Box 7.1](#).

Nephrotic Syndrome

Routine clinical and serologic examination of patients with nephrotic syndrome usually allows the clinician to determine whether a systemic disorder is present. In adults and in adolescents beyond puberty without systemic disease, the glomerular pathologic process cannot be predicted with confidence by noninvasive criteria alone; therefore a renal biopsy should be performed. In children age 1 year up to puberty, a presumptive diagnosis of minimal change disease (MCD) usually can be made. Kidney biopsy is reserved for nephrotic children with atypical features, including microhematuria, hypocomplementemia, reduced glomerular filtration rate (GFR), and failure to respond to corticosteroids. Antibodies against phospholipase A2 receptor 1 (PLA₂R1) are found in 80% of patients with membranous nephropathy and are a noninvasive aid to diagnosis,⁴ response to treatment,⁵ and prediction of remission.⁶ Unfortunately, the test only identifies 70% of cases of primary membranous nephropathy. In addition, biopsy still provides information on disease chronicity and the extent of irreversible fibrosis.

Acute Kidney Injury

In most patients with acute kidney injury (AKI) on a background of chronic kidney disease (CKD), the cause can be determined without a renal biopsy. In a minority of patients, however, a confident diagnosis

cannot be made, and a renal biopsy should be performed urgently so appropriate treatment can be started before irreversible renal injury develops. This is particularly true in patients with AKI accompanied by an active urine sediment or with suspected drug- or infection-induced acute interstitial nephritis.

Systemic Disease Associated With Kidney Dysfunction

Patients with diabetes mellitus and kidney dysfunction do not usually require biopsy if they have clinical features that suggest diabetic nephropathy (e.g., isolated proteinuria, diabetes of long duration, other microvascular complications). However, kidney biopsy should be performed if the presentation is atypical, such as proteinuria associated with glomerular hematuria, absence of retinopathy or neuropathy (in patients with type 1 diabetes), onset of proteinuria less than 5 years from the documented onset of diabetes, uncharacteristically rapid change in kidney function or kidney disease of acute onset, or immunologic abnormalities.

Serologic testing for antineutrophil cytoplasmic antibody (ANCA) and for anti-glomerular basement membrane (anti-GBM) antibodies usually allows a confident diagnosis of kidney small-vessel vasculitis or Goodpasture disease without invasive measures. Nonetheless, a kidney biopsy still should be performed to confirm the diagnosis and clarify the extent of chronic fibrosis and thus the potential for recovery. This information may be important in helping decide whether to initiate or continue immunosuppressive therapy, particularly when complications of immunosuppression are observed or expected.

Lupus nephritis usually can be diagnosed by noninvasive criteria such as autoantibodies, urine protein excretion, kidney function, and urine sediment abnormalities, which can also be used to inform decisions about initial immunosuppressive treatment. However, a kidney biopsy will clarify the underlying pathologic lesion, level of active inflammation, and extent of chronic fibrosis, thereby providing robust guidance for therapy.

The diagnosis of viral infection-related nephropathy (e.g., hepatitis B virus-associated membranous nephropathy) is suggested by characteristic histologic lesions together with evidence of active viral infection. However, the identification of virus-specific protein or DNA or RNA in the kidney biopsy tissue by immunopathologic and molecular pathologic techniques (e.g., in situ hybridization) can confirm the diagnosis.

Other systemic diseases, such as amyloidosis, sarcoidosis, and myeloma can be diagnosed with kidney biopsy. However, a kidney biopsy is indicated only if the diagnosis remains uncertain or if knowledge of kidney involvement would change management.

Kidney Transplant Dysfunction

Kidney levels of calcineurin inhibitors require a kidney biopsy to determine the cause. In the early posttransplantation period, this is most

BOX 7.1 Indications for Kidney Biopsy**Nephrotic Syndrome**

- Routinely indicated in adults
- In prepubertal children, indicated only if clinical features atypical of minimal change disease are present

Acute Kidney Injury

- Indicated if obstruction, reduced kidney perfusion, and acute tubular necrosis have been ruled out

Systemic Disease With Kidney Dysfunction

- Indicated in patients with small-vessel vasculitis, anti-glomerular basement membrane disease, and systemic lupus
- Indicated in patients with diabetes only if atypical features present

Nonnephrotic Proteinuria

- May be indicated if proteinuria is >1 g/24 h

Isolated Microscopic Hematuria

- Indicated only in unusual circumstances

Unexplained Chronic Kidney Disease

- May be diagnostic (e.g., identify immunoglobulin A nephropathy even in “end-stage kidney”)

Familial Kidney Disease

- Biopsy of one affected member may give diagnosis and minimize further investigation of family members

Kidney Transplant Dysfunction

- Indicated if ureteral obstruction, urinary sepsis, renal artery stenosis, and toxic calcineurin inhibitor levels are not present

useful in differentiating acute rejection from acute tubular necrosis (ATN) and the BK virus nephropathy. Later, kidney biopsy can differentiate acute rejection from chronic allograft nephropathy, recurrent or de novo glomerulonephritis (GN), and calcineurin inhibitor toxicity. The location of the kidney transplant in the iliac fossa facilitates biopsy of the allograft and allows repeated biopsies when indicated. This has encouraged many units to adopt a policy of protocol (surveillance) biopsies to detect subclinical acute rejection and kidney scarring and to guide the choice of immunosuppressive therapy (see [Chapter 109](#)).

Nonnephrotic Proteinuria

The value of kidney biopsy in patients with nonnephrotic proteinuria is debatable. All conditions that result in nephrotic syndrome can cause nonnephrotic proteinuria, except for MCD. However, the benefit of specific treatment with corticosteroids and other immunosuppressive agents in these patients probably does not justify the risk for significant drug-related side effects. In patients with proteinuria of more than 1 g/day, generic treatment with strict blood pressure control and angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs) reduces proteinuria and reduces the risk for development of progressive kidney dysfunction (see [Chapter 82](#)). Nonetheless, although the kidney biopsy may not lead to an immediate change in management, it can be justified in these circumstances if it will provide prognostic information, identify a disease for which a different therapeutic approach is indicated, or provide clinically important information about the future risk for disease recurrence after kidney transplantation.

Isolated Microhematuria

Patients with microhematuria should be evaluated to identify structural lesions such as kidney stones or kidney and urothelial malignant neoplasms. The absence of a structural lesion suggests that the hematuria may have a glomerular source. Biopsy studies have identified glomerular lesions in up to 75% of biopsies.⁷ In all series, immunoglobulin A (IgA) nephropathy is the most common lesion, followed by thin basement membrane nephropathy and normal kidney morphology. In the absence of nephrotic proteinuria, kidney impairment, or hypertension, the prognosis for patients with these conditions is excellent, and because specific therapies are not available, kidney biopsy is not necessary and patients require only follow-up. Biopsy should be performed only if the result would reassure the patient, avoid repeated urologic investigations, or provide specific information, as in the evaluation of potential living kidney donors, in familial hematuria, or for life insurance and employment purposes.

Unexplained Chronic Kidney Disease

Kidney biopsy can be informative in the patient with unexplained chronic kidney impairment and normal-sized kidneys, because in contrast to AKI, it is often difficult to determine the underlying cause with clinical criteria alone. In almost half of such patients, the biopsy will demonstrate an unexpected cause of CKD.⁸ However, if both kidneys are small (<9 cm on ultrasound), the risks of biopsy are increased, and the diagnostic information may be limited by extensive glomerulosclerosis and tubulointerstitial fibrosis. In this setting, however, immunofluorescence studies still may be informative. For example, glomerular IgA deposition may be identified despite advanced structural damage.

Familial Kidney Disease

A kidney biopsy can be helpful in the investigation of patients with a family history of kidney disease. A biopsy performed in one affected family member may secure the diagnosis for the whole family and avoid the need for repeat investigation. Conversely, a kidney biopsy may unexpectedly identify inherited disease, thereby prompting evaluation of other family members.

Role of Repeat Kidney Biopsy

In some patients, a repeat biopsy may be indicated. For example, the pathologic changes in lupus nephritis may evolve, necessitating treatment adjustment. In addition, corticosteroid-resistant, corticosteroid-dependent, or frequently relapsing MCD may represent a missed diagnosis of focal segmental glomerulosclerosis (FSGS), which may be detected on repeat biopsy. Some nephrologists think repeat biopsy in patients who have had aggressive immunosuppressive therapy of crescentic GN can help determine the most appropriate next line of therapy.

VALUE OF KIDNEY BIOPSY**Biopsy Adequacy**

In the assessment of a kidney biopsy, the number of glomeruli in the sample is the major determinant of whether the biopsy will be diagnostically informative.

For a focal entity such as FSGS, the diagnosis could be made on a biopsy specimen containing a single glomerulus that contains a typical sclerosing lesion. However, the probability that FSGS is not present in a patient with nephrotic syndrome and minimal changes on the biopsy specimen depends on the actual proportion of abnormal glomeruli in the kidney and the number of glomeruli obtained in the biopsy specimen. For example, if 20% of glomeruli in the kidney have sclerosing

Effect of Sampling on Kidney Biopsy Interpretation

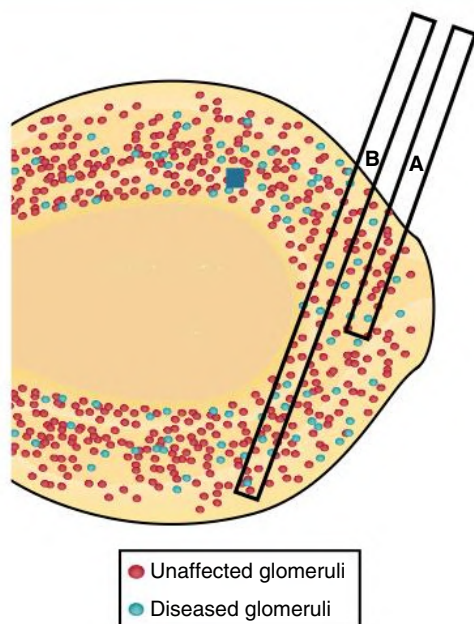


Fig. 7.1 Effect of Sampling on Kidney Biopsy Interpretation. Red dots represent unaffected glomeruli. Blue dots represent diseased glomeruli. The size of the biopsy core affects the probability that the observed glomerular involvement is a true reflection of involvement in the whole kidney. In a biopsy specimen containing 10 glomeruli (core A), of which three are abnormal (30%), there is a 95% probability that the actual glomerular involvement is between 7% and 65%. In the same kidney, if the biopsy specimen contained 30 glomeruli with 30% being abnormal (core B), the 95% confidence intervals are narrowed to 15% and 50%.

lesions and five glomeruli are sampled, there is a 35% chance that all the glomeruli in the biopsy specimen will be normal and the biopsy will miss the diagnosis. By contrast, in the same kidney, if 10 or 20 glomeruli are sampled, the chance of obtaining all normal glomeruli is reduced to 10% and less than 1%, respectively, and the biopsy is therefore more discriminating. This argument assumes that any segmental lesions present in the biopsy specimen are actually identified; this requires the biopsy specimen to be sectioned at multiple levels.

Unless all glomeruli are affected equally, the probability that the observed involvement in the biopsy specimen accurately reflects true involvement in the kidney depends not only on the number of glomeruli sampled but also on the proportion of affected glomeruli (Fig. 7.1).

Therefore the interpretation of the biopsy should account for the number of glomeruli obtained. A typical biopsy sample will contain 10 to 15 glomeruli and will be diagnostically useful. Nonetheless, it must be appreciated that because of the sampling issue, a biopsy sample of this size will occasionally be unable to diagnose focal diseases and at best will provide imprecise guidance on the extent of glomerular involvement.

An adequate biopsy also should provide samples for immunohistologic examination and electron microscopy (EM). Immunohistologic examination is performed by either immunofluorescence on frozen material or immunoperoxidase on fixed tissue, according to local protocols and expertise. It is helpful for the biopsy cores to be viewed under an operating microscope immediately after being taken to ensure that they contain cortex and that when the cores are divided, the immunohistologic and EM samples both contain glomeruli.

If the material obtained for a complete pathologic evaluation is insufficient, a discussion with the pathologist should address how best

to proceed before the tissue is placed in fixative, so the material can be processed in a way that will provide maximum information for the specific clinical scenario. For example, if the patient has heavy proteinuria, most information will be gained from EM because it can demonstrate podocyte foot process effacement, focal sclerosis, electron-dense deposits of immune complexes, and the organized deposits of amyloid.

If a sample is supplied for immunofluorescence microscopy but contains no glomeruli, it may be possible to reprocess the paraffin-embedded part of the sample to identify immune deposits by immunoperoxidase or immunofluorescence techniques.

Is Kidney Biopsy Always Necessary?

Early studies suggested that kidney biopsy provided diagnostic clarity in most patients, but that this information did not alter management, except for those with heavy proteinuria or systemic disease. More recent prospective studies have suggested that the kidney biopsy identifies a diagnosis different from that predicted on clinical grounds in 50% to 60% of patients and leads to a treatment change in 20% to 50%.⁹ This is particularly apparent in patients with heavy proteinuria or AKI, more than 80% of whom have biopsy findings that alter their management.¹⁰

PREBIOPSY EVALUATION

The prebiopsy evaluation identifies issues that may compromise the safety and success of the procedure (Fig. 7.2). It will determine whether the patient has two normal-sized unobstructed kidneys, sterile urine, controlled blood pressure, and no bleeding diathesis. A thorough history should be taken to identify evidence of a bleeding diathesis, such as previous prolonged surgical bleeding, spontaneous bleeding, family history of bleeding, and ingestion of medication that increases bleeding risk, including antiplatelet agents and anticoagulants.

Important clinical history also relates to patients' previous experiences of biopsies as well as psychological conditions and level of comprehension.¹¹

An ultrasound scan should be performed to assess kidney size and identify significant anatomic abnormalities, such as solitary kidney, polycystic or simple cystic kidneys, malpositioned kidneys, horseshoe kidneys, small kidneys, and hydronephrosis.

We recommend hematology studies prebiopsy including platelet count and coagulation profile with prothrombin time and activated partial thromboplastin time. The value of the bleeding time (BT) and thromboelastography (TEG) for predicting bleeding risk after kidney biopsy is unclear. Retrospective studies, however, demonstrated a three- to fivefold increase in bleeding complications after kidney biopsy in patients with prolonged BT. Prospective studies of percutaneous liver biopsy patients showed a fivefold increase in bleeding complications in those with uncorrected BT.¹² A consensus document concluded that the BT is a poor predictor of postsurgical bleeding, but it does correlate with clinical bleeding episodes in uremic patients.¹³

TEG provides an overall measure of the coagulation, platelet, and fibrinolytic systems in one assay and thus may be more predictive of clinical bleeding.¹⁴ A more recent study of 417 patients indicated that TEG abnormalities in patients with renal dysfunction are variable and fail to predict bleeding after kidney biopsy.¹⁵ The role of TEG in the patient undergoing native kidney biopsy requires further evaluation.

Management of Bleeding Risk

Several approaches to the management of bleeding risk have been adopted.

Aspirin

Patients should ideally discontinue aspirin 7 days prior to kidney biopsy. However, because cardiovascular disease is very common in

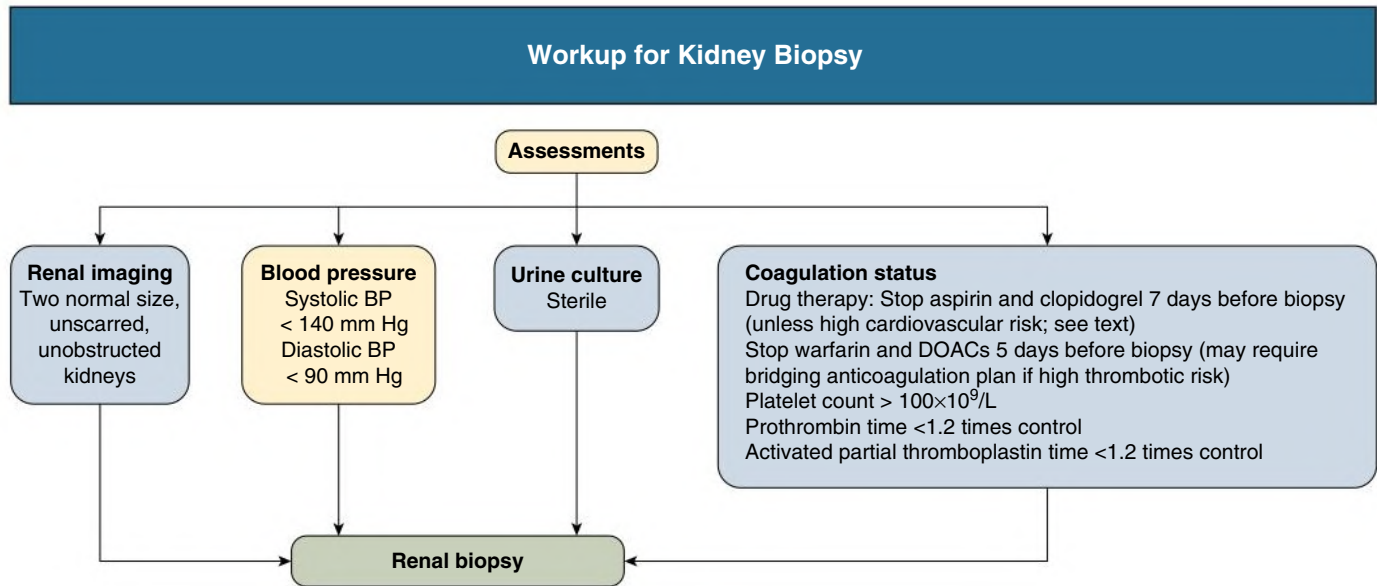


Fig. 7.2 Workup for Kidney Biopsy. BP, Blood pressure; DOACs, direct oral anticoagulants.

subjects with kidney disease, aspirin discontinuation policies have led to increased cardiac events of recurrent myocardial infarction of 40% and stroke increase three- to fourfold.¹⁶ Although there is an associated increased risk of minor bleeding but not major bleeding, aspirin should be continued in the setting of active cardiovascular risk (e.g., ongoing angina, recent coronary angioplasty, transient ischemic attack).¹⁷

Clopidogrel

There is very limited data examining safety of continuing clopidogrel at the time of biopsy, so most guidelines suggest cessation 7 days prior to biopsy. Platelet transfusion also can be used to reverse clopidogrel-induced platelet dysfunction when the kidney biopsy is urgent.

Warfarin and Direct Oral Anticoagulants

Warfarin and direct oral anticoagulants (DOACs) should be stopped 5 days prior to biopsy. In low-risk patients, there is no need to have bridging anticoagulation,¹⁸ but in individuals with a mechanical valve, antiphospholipid syndrome, high-risk thrombophilia, or venous thromboembolism within 3 months, bridging low-molecular-weight or unfractionated heparin should be used.¹⁷

Procoagulants

The use of desmopressin (DDAVP) has been suggested in a small study with minimal clinically significant effect¹⁹ as a strategy to lessen minor and major bleeding because it is a synthetic analog of vasopressin and increases platelet adhesion to the blood vessel wall by releasing factor VII and von Willebrand factor. Because of its increased side effect profile (hyponatremia, hypotension, tachycardia, facial flushing, nausea, and abdominal pain) and minimal clinical evidence supporting its use, it is no longer recommended.¹⁷

Contraindications to Kidney Biopsy

The contraindications to percutaneous kidney biopsy are listed in Box 7.2. The major contraindication is a bleeding diathesis. If the disorder cannot be corrected and the biopsy is deemed indispensable, alternative approaches can be used, such as open biopsy, laparoscopic biopsy, or transvenous (usually transjugular) biopsy. Inability of the patient to comply with instructions during kidney biopsy is another major contraindication. Sedation or, in extreme cases, general anesthesia may be necessary.

BOX 7.2 Contraindications to Kidney Biopsy

Kidney Status

- Multiple cysts
- Solitary kidney
- Acute pyelonephritis
- Perinephric abscess
- Kidney neoplasm

Patient Status

- Uncontrolled bleeding diathesis
- Uncontrolled blood pressure
- Uremia
- Obesity
- Uncooperative patient

Most contraindications are relative rather than absolute. Clinical circumstances that necessitate urgent kidney biopsy may be overridden, except for uncontrolled bleeding diathesis.

Hypertension (>140/90 mm Hg), hypotension, perinephric abscess, pyelonephritis, hydronephrosis, severe anemia, large kidney tumors, and cysts are relative contraindications to kidney biopsy. When possible, these should be corrected before the biopsy is undertaken.

The presence of a solitary functioning kidney has been considered a contraindication to percutaneous biopsy, and some argue that the risk of biopsy is reduced by direct visualization at open biopsy. However, the postbiopsy nephrectomy rate of 1 in 2000 to 1 in 5000 is comparable to the mortality rate associated with the general anesthetic required for an open procedure. Therefore in the absence of risk factors for bleeding, percutaneous biopsy of a solitary functioning kidney can be justified if there is a clinically important indication for kidney biopsy.

KIDNEY BIOPSY TECHNIQUE

Percutaneous Kidney Biopsy

Native Kidney Biopsy

At our centers, the kidney biopsy is performed by nephrologists with continuous (real-time) ultrasound guidance and disposable automated biopsy needles. We use 16-gauge needles as a compromise between the greater tissue yield of larger needles and the trend toward fewer bleeding

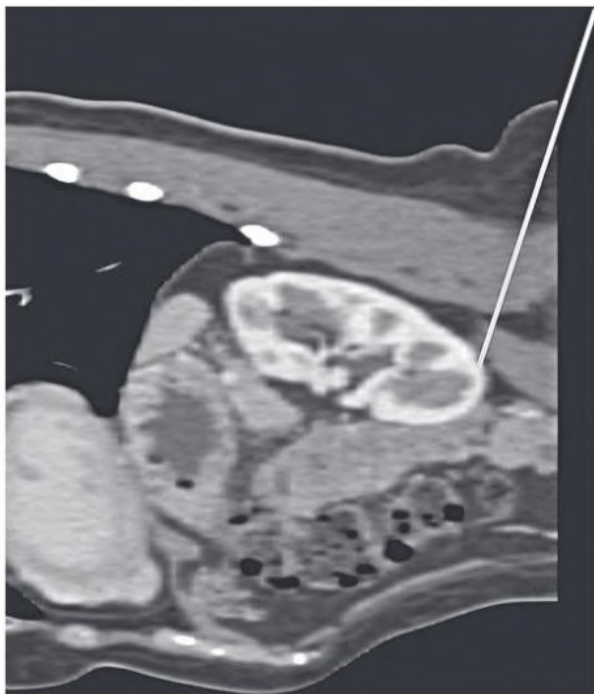


Fig. 7.3 Computed Tomography Through the Left Kidney. The angle of approach of the needle is demonstrated. Note the relative adjacency of the lower pole of the kidney to other structures, particularly the large bowel.

complications of smaller needles as recommended by recent guidelines.¹⁷ For most patients, premedication or sedation is not required as long as there is adequate education and informed consent provided. A history of anxiety may require prebiopsy counseling and intra-procedural relation therapy.¹¹ The patient is prone, and a pillow is placed under the abdomen at the level of the umbilicus to straighten the lumbar spine and splint the kidneys. **Fig. 7.3** shows the anatomic relationships of the left kidney. Ultrasound is used to localize the lower pole of the kidney where the biopsy will be performed (usually the left kidney). An indelible pen mark is used to indicate the point of entry of the biopsy needle. The skin is sterilized with povidone-iodine (Betadine) or chlorhexidine solution. A sterile fenestrated sheet is placed over the area to maintain a sterile field. Local anesthetic (2% lidocaine [lignocaine]) is infiltrated into the skin at the point previously marked.

While the anesthetic takes effect, the ultrasound probe is covered in a sterile sheath. Sterile ultrasound jelly is applied to the skin and, under ultrasound guidance, a 10-cm, 21-gauge needle is guided to the renal capsule and further local anesthetic infiltrated into the perirenal tissues, then along the track of the needle on withdrawal. A stab incision is made through the dermis to ease passage of the biopsy needle. This is passed under ultrasound guidance to the kidney capsule (**Fig. 7.4**). As the needle approaches the capsule, the patient is instructed to take a breath until the kidney is moved to a position such that the lower pole rests just under the biopsy needle, and then to stop breathing. The biopsy needle tip is advanced to the kidney capsule, and the trigger mechanism is released, firing the needle into the kidney (**Fig. 7.5**). The needle is immediately withdrawn, the patient is asked to resume breathing, and the contents of the needle are examined (**Fig. 7.6**). We examined the tissue core under an operating microscope to ensure that kidney cortex has been obtained (**Fig. 7.7**). A second pass of the needle is usually necessary to obtain additional tissue for immunohistologic examination and EM. If insufficient tissue is obtained, further passes of the needle are made. In our experience, however, passing the needle more than four times is associated with a modest increase in the postbiopsy complication rate.



Fig. 7.4 Kidney Biopsy Procedure. The biopsy needle is introduced at an angle of approximately 70 degrees to the skin and is guided by continuous ultrasound. The operator is shown wearing a surgical gown. This is not strictly necessary; sterile gloves and maintenance of a sterile field are sufficient.



Fig. 7.5 Kidney Biopsy Imaging. Ultrasound scan shows the needle entering the lower pole of the left kidney. Arrows indicate the needle track, which appears as a fuzzy white line.

No single fixative has been developed that allows good-quality light microscopy, immunofluorescence, and EM to be performed on the same sample. Therefore the kidney tissue is often divided into three samples and placed in formalin for light microscopy, normal saline for subsequent snap-freezing in liquid nitrogen for immunofluorescence, and glutaraldehyde for EM. Some centers can produce satisfactory light microscopy, immunohistochemistry, and EM on formalin-fixed biopsy material, although this depends on the expertise of individual laboratories.

The percutaneous kidney biopsy technique has several variations. Whereas most biopsies are guided by ultrasound, some operators use ultrasound only to localize the kidney and determine the depth and angle of approach of the needle and perform the biopsy without further ultrasound guidance. The success and complication rates appear

to be no different from those seen with continuous ultrasound guidance. For technically challenging biopsies, computed tomography (CT) can be used to guide the biopsy needle.

For obese patients and patients with respiratory conditions who find the prone position difficult, the supine anterolateral approach recently has been described.²⁰ This technique provides good access to the lower pole of the kidney, is better tolerated than the prone position by these patients, and has a diagnostic yield and safety profile comparable to that of the standard technique for native kidney biopsy.

Kidney Transplant Biopsy

Biopsy of the transplant kidney is facilitated by the proximity of the kidney to the anterior abdominal wall and the lack of movement on respiration. It is performed under real-time ultrasound guidance with use of an automated biopsy needle. It is important to confirm that the transplanted kidney is in the normal extraperitoneal position—occasionally it will be intraperitoneal (simultaneous pancreas and kidney transplants in particular), and bowel injury becomes a potential hazard. In most patients, the kidney transplant biopsy is performed to identify the cause of acute allograft dysfunction. In these circumstances, the goal is to identify acute rejection, and therefore the diagnosis can be made on a formalin-fixed sample alone for light microscopy. If vascular rejection is suspected, a snap-frozen sample for C4d immunostaining also should be obtained (although some laboratories can detect C4d on formalin-fixed material). If recurrent or de novo GN is suspected in

patients with chronic allograft dysfunction, additional samples for EM and immunohistologic examination should be collected.

Special Situations

Pregnancy. In the setting of pregnancy, the risk of biopsy must be weighed against any potential benefit of diagnosis and treatment, especially considering that treatment options may be limited due to teratogenicity. Biopsy has been shown safe in all stages of pregnancy, but often after 32 weeks the baby is delivered first. A lateral position is undertaken to be able to perform the biopsy safely. A recent review study in which biopsy was performed for suspected glomerulonephritis or preeclampsia reported that therapy was altered in 66% of cases, suggesting it is useful in selected cases.²¹

Pediatrics. Renal biopsy in the pediatric setting has the added difficulty of much smaller kidneys and the need for a general anesthetic due to patient cooperation. Multiple cohort studies report that ultrasound-guided, percutaneous biopsies with a spring-loaded device and 18-gauge needle are safe with a complication rate of 4.1% and are successful in 99% of cases,²² thus with a reduced number of passes resulting in similar complication rates. Kidney biopsy in pediatric kidney transplantation for surveillance has also been shown to have the same safety profile as in adults.²³

Postbiopsy Monitoring

After the biopsy, the patient is placed supine and subjected to strict bed rest for 6 to 8 hours. The blood pressure is monitored frequently, the urine examined for visible hematuria, and the skin puncture site examined for excessive bleeding. If there is no evidence of bleeding after 6 hours, the patient is sat up in bed and subsequently allowed to ambulate. If visible hematuria develops, bed rest is continued until the bleeding settles. We advise minimal activity for 48 hours after biopsy and avoidance of contact sports and activities requiring straining for 2 weeks.

Conventionally, patients have been observed for complications in the hospital for 24 hours after biopsy. However, outpatient kidney biopsy with same-day discharge after 6 to 12 hours of observation has become increasingly popular for both native and kidney transplant biopsies. We recommend 6 hours for low-risk and 12 hours for high-risk biopsies (patients with significant renal impairment, patients undergoing biopsy for AKI, patients with elevated blood pressure prebiopsy, patients aged more than 70 years, and patients with abnormal bleeding profile or requiring early recommencement of

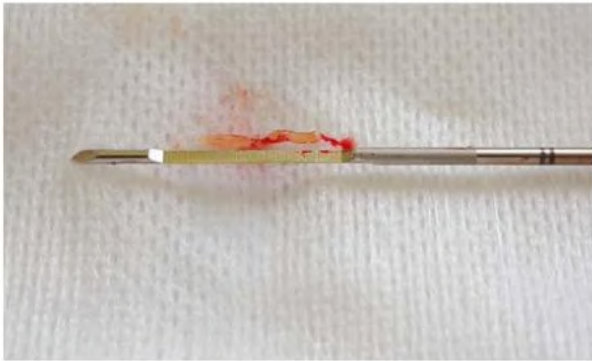


Fig. 7.6 Kidney Biopsy Sample. A core of kidney tissue is demonstrated in the sampling notch of the biopsy needle.

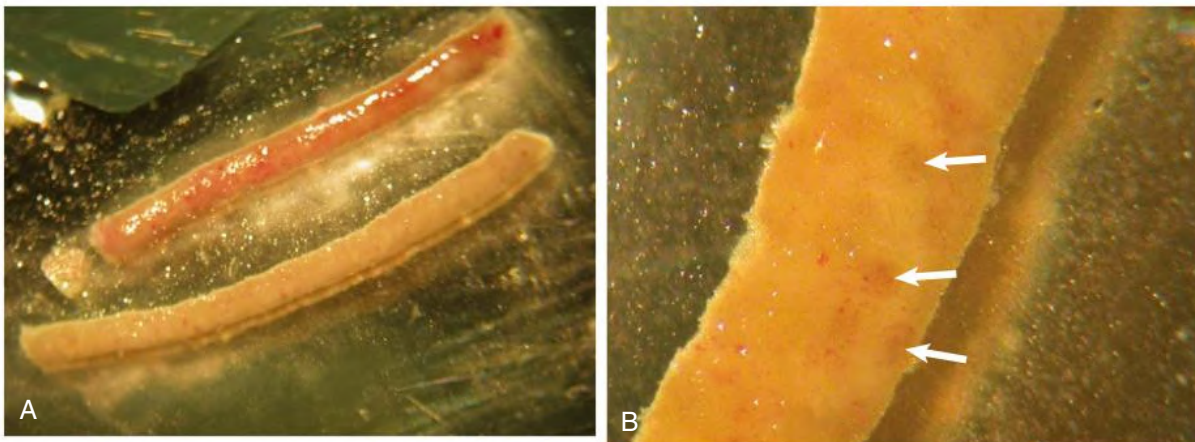


Fig. 7.7 Kidney Biopsy Micrographs. Appearance of kidney biopsy material under the operating microscope. (A) Low-power view shows two good-sized cores. (B) Higher magnification view shows the typical appearance of glomeruli (arrows).

anticoagulation).¹⁷ Some studies suggest that as much as one-third of serious bleeding complications requiring transfusion may occur after 8 hours postbiopsy, whereas up to 91% occur within 12 hours.²⁴

Early identification of postbiopsy bleeding is potentially useful in patients with stable observations and no clinical symptoms of bleeding. Ultrasound scanning 1 hour after biopsy has been investigated as a method to predict bleeding complications.^{25,26} In the older study the absence of hematoma predicted an uncomplicated course, but identifying a hematoma did not reliably predict adverse outcomes; identification of hematoma at 1 hour had a 95% negative predictive value and 43% positive predictive value for a significant complication. The role of postbiopsy imaging in the wider clinical setting remains to be determined given the additional expense of routine scanning and the uncertainty about how best to manage findings that are detected only by ultrasound—an issue that was not addressed in the more recent study.²⁶

Alternatives to the Percutaneous Approach

When the percutaneous approach is contraindicated, other approaches to kidney biopsy have been described. The choice of technique depends on the safety, morbidity, recovery period, and adequacy of the technique, but mainly on the local expertise available.

Transvenous (Transjugular or Transfemoral) Kidney Biopsy

Transvenous sampling of the kidney is theoretically safer than the percutaneous approach because the needle passes from the venous system into the renal parenchyma and is directed away from large blood vessels. Any bleeding that occurs should be directed back into the venous system, and if capsular perforation develops, significant bleeding points can be immediately identified and controlled by coil embolization. Others argue that coil embolization of the punctured vein is unhelpful because significant bleeding into a perirenal hematoma or the urine indicates an arterial breach that requires selective angiography and arterial embolization.

Transvenous kidney biopsy cannot be regarded as routine because it involves specialist skills and additional time and expense compared with the percutaneous approach. The main indication for this approach is an uncontrollable bleeding diathesis. It also has been advocated for patients receiving artificial ventilation in the intensive care unit; the need to obtain tissue from more than one organ, including the kidney, liver, or heart; large-volume ascites that preclude the prone position; uncontrolled hypertension; morbid obesity; severe respiratory insufficiency; solitary kidney; failed percutaneous approach; and coma.

The patient lies supine, and the right internal jugular vein is cannulated. A guidewire is passed into the inferior vena cava (IVC), and a catheter is passed over the guidewire and selectively into the right renal vein, which is shorter and enters the IVC at a more favorable angle than the left renal vein. A sheath is passed over the catheter to a suitable peripheral location in the kidney with the aid of contrast enhancement. Finally, the biopsy device (usually a side-cut biopsy needle system) is passed through the sheath, and samples are taken. Contrast is then injected into the biopsy track to identify capsular perforation, and embolization coils are inserted if brisk bleeding is identified.

The quality of kidney tissue obtained by transjugular biopsy is variable, although studies report diagnostic yields of more than 90%.²⁷ The complication rate appears comparable to that seen with percutaneous kidney biopsy, which is reassuring given that these are high-risk patients.

Open Kidney Biopsy

Open kidney biopsy is useful when percutaneous biopsy is contraindicated. In a series of 934 patients, tissue adequacy was 100% with no major complications.²⁸ This is an effective approach with minimal

postprocedure complications, but the risk of general anesthesia and the delayed recovery time have prevented its widespread adoption. Open biopsy still may be performed, however, when a kidney biopsy is required in patients who are otherwise undergoing abdominal surgery.

Laparoscopic Kidney Biopsy

Laparoscopic renal biopsy requires general anesthesia and two laparoscopic ports in the posterior and anterior axillary lines to gain access to the retroperitoneal space. Laparoscopic biopsy forceps are used to obtain cortical biopsy samples, and the biopsy sites are coagulated with laser and packed to prevent hemorrhage. In the largest study of laparoscopic kidney biopsy, adequate tissue was obtained in 96% of 74 patients.²⁹ Significant bleeding occurred in three patients, the colon was injured in one, and a biopsy was performed inadvertently on the spleen and liver, respectively, in two others. Inadvertent biopsy was subsequently averted using intraoperative ultrasound in difficult cases.

COMPLICATIONS OF KIDNEY BIOPSY

The complication rates compiled from large series of kidney biopsies are shown in Table 7.1.³⁰

Pain

Patients should be informed about the inevitable dull ache around the needle entry site when the local anesthetic wears off after renal biopsy. Simple analgesia with acetaminophen (paracetamol) or acetaminophen-codeine combinations usually suffices. More severe pain in the loin or abdomen on the side of the biopsy suggests significant perirenal hemorrhage. Opiates may be necessary for pain relief, with appropriate investigation to clarify the severity of the bleed. Patients with visible hematuria may develop clot colic and describe the typical severe pain associated with ureteral obstruction.

Hemorrhage

A degree of perirenal bleeding accompanies every kidney biopsy. The mean decrease in hemoglobin after a biopsy is approximately 1 g/dL.³¹ Significant perirenal hematomas are almost invariably associated with severe loin pain, although some hematomas are asymptomatic. Both visible hematuria and painful hematoma are seen in 3% to 4% of patients after biopsy.

The initial management is strict bed rest and maintenance of normal coagulation indices. If bleeding is brisk and associated with hypotension or prolonged and fails to settle with bed rest, renal angiography should be performed to identify the source of bleeding. Coil

TABLE 7.1 Complications in 118,064 Native Kidney Biopsies

Complication	Percentage
Hematoma	11
Pain	4.3
Visible hematuria	3.5
Need for blood transfusion	1.6
Need for intervention to control bleeding	0.3
Death	0.06 (0.03 for outpatient biopsies)

Data from Douketis JD, Spyropoulos AC, Kaatz S, et al. Perioperative bridging anticoagulation in patients with atrial fibrillation. *N Engl J Med*. 2015;373(9):823–833.

embolization can be performed during the same procedure, and this has largely eliminated the need for open surgical intervention and nephrectomy.

Arteriovenous Fistula

Most postbiopsy arteriovenous fistulas are detected by Doppler ultrasound or contrast-enhanced CT and, when looked for specifically, can be found in as many as 18% of patients. Because most are clinically silent and more than 95% resolve spontaneously within 2 years, fistulas should not be routinely sought. In a small minority of patients, arteriovenous fistulas can lead to visible hematuria (typically recurrent, dark red, and often with blood clots), hypertension, and renal impairment, which requires embolization.

SELF-ASSESSMENT QUESTIONS

1. A 68-year-old woman with nonnephrotic proteinuria of unknown cause attended the outpatient unit for a kidney biopsy. She had a history of hypertension, chronic alcohol abuse, and osteoarthritis. Which of the following would be a contraindication to renal biopsy for this patient?
 - A. Ingestion of naproxen 24 hours earlier
 - B. Blood pressure of 162/94 mm Hg
 - C. Dipstick urinalysis positive for nitrites and leukocytes
 - D. Presence of simple cysts in both kidneys (four in left, three in right)
 - E. Body mass index of 36 kg/m²
2. A 32-year-old woman was referred to the Kidney-obstetric clinic for evaluation of kidney disease. Which is a recognized indication for kidney biopsy during pregnancy?
 - A. Isolated proteinuria, ratio of protein to creatinine of 320 mg/mmol
 - B. Asymptomatic proteinuria with nonvisible hematuria
 - C. Episodic visible hematuria
 - D. Symptomatic nephrotic syndrome after 32 weeks of gestation
 - E. Unexplained deterioration in kidney function before 32 weeks of gestation
3. A 75-year-old man was admitted as an emergency with AKI. He had been unwell for 4 months with myalgia, arthralgia, and generalized fatigue. More recently, he had developed epistaxis and a nonblanching leg rash. Six months earlier, he had a myocardial infarction followed by right coronary artery angioplasty and stenting. He was taking aspirin (75 mg), clopidogrel (75 mg), bisoprolol (5 mg), ramipril (5 mg), and atorvastatin (40 mg). On examination, his blood pressure was 148/86 mm Hg, and he was euvoletic. A purpuric rash was present on both lower legs. Test results were as follows:
 - Serum creatinine: 523 µmol/L (60–110)

Other Complications

A variety of rare complications have been reported, including biopsy performed on other organs (liver, spleen, pancreas, bowel, gallbladder), pneumothorax, hemothorax, calyceal-peritoneal fistula, dispersion of carcinoma, and Page kidney (compression of kidney by perirenal hematoma leading to renin-mediated hypertension).

Death

Death resulting directly from kidney biopsy is much less common in recent biopsy series compared with earlier reports. Most deaths are the result of uncontrolled hemorrhage in high-risk patients, particularly those with severe kidney impairment.

- Serum C-reactive protein: 25 mg/L (<10)
- Urine dipstick: 3+ blood, 2+ protein, 0 nitrites, 0 leukocytes
- Antinuclear antibodies: negative
- Anti-neutrophil cytoplasmic antibodies: negative
- Anti-glomerular basement membrane antibodies: negative

An urgent kidney biopsy is requested to determine the cause of AKI. What should be done to minimize the risk for postbiopsy bleeding?

- A. Undertake a transjugular kidney biopsy
 - B. Transfuse platelets before the biopsy
 - C. Administer DDAVP
 - D. Stop aspirin and clopidogrel and wait 24 hours before performing the biopsy
 - E. Administer vitamin K and proceed if international normalized ratio and activated partial thromboplastin time ratio are normal
4. A 32-year-old man was seen in the nephrology clinic with proteinuria. He has a history of type 1 diabetes mellitus and hypertension. He was treated with lisinopril (10 mg), amlodipine (5 mg), aspirin (75 mg), simvastatin (40 mg), and insulin. On examination, he was overweight (body mass index: 30 kg/m²), blood pressure was 146/84 mm Hg, and edema was present to midcalf level bilaterally. Test results were as follows:
 - Serum creatinine: 123 µmol/L (60–110)
 - Serum albumin: 28 g/L (37–49)
 - Urinary protein/creatinine ratio: 460 mg/mmol (<15)
 Which additional feature would provide justification for undertaking a kidney biopsy?
 - A. Presence of diabetic retinopathy
 - B. Duration of diabetes of 12 years
 - C. Presence of nonselective proteinuria
 - D. Negative dipstick urinalysis 4 months earlier
 - E. Hemoglobin A1c of 7.2%

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